Akoizo Original Practice - Form 1 - Answer key

Division C | Rocks And Minerals | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Mohs hardness is…

Unit: Minerals, rocks, and diagnostic evidence.
Worked-case data: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

A. An ordinal scratch scale

B. A linear mass scale

C. A color scale

D. A toughness scale

Correct: A

2. (2 points) Equal axes and right angles describe the conventional…

Unit: Formation environments, crystal systems, and habit.
Worked-case data: A massive specimen lacks visible faces. Does it lack crystalline order?

A. Monoclinic system

B. Orthorhombic system

C. Cubic system

D. Triclinic system

Correct: C

3. (2 points) Micas are representative…

Unit: Silicate structures and mineral groups.
Worked-case data: Why does a fully shared silica framework have SiO_2 rather than SiO_4?

A. Sheet silicates

B. Native elements

C. Halides

D. Sulfides

Correct: A

4. (2 points) Pyrite belongs to…

Unit: Non-silicates and economic geology.
Worked-case data: A dark specimen has reddish-brown streak and weak magnetism. Which iron mineral is favored?

A. Carbonates

B. Native elements

C. Sulfides

D. Halides

Correct: C

5. (2 points) A two-stage cooling clue is…

Unit: Igneous processes, textures, and Bowen's series.
Worked-case data: Granite and rhyolite have similar broad composition but different textures. Explain.

A. Porphyritic texture

B. One uniform streak

C. Metallic luster

D. Perfect sorting

Correct: A

6. (2 points) Angular gravel clasts suggest…

Unit: Sedimentary processes, grains, and fossils.
Worked-case data: A rock contains rounded gravel clasts in cement. Classify and constrain the claim.

A. Obsidian

B. Schist

C. Breccia

D. Conglomerate specifically

Correct: C

7. (2 points) Metamorphism usually involves…

Unit: Metamorphism, stress, and mineral assemblages.
Worked-case data: A carbonate-rich rock near an intrusion recrystallizes without melting. Propose an interpretation.

A. Solid-state change

B. Complete melting by definition

C. Only loose sediment

D. No chemical reactions

Correct: A

8. (2 points) Best reference design includes…

Unit: Identification strategy, references, and timed practice.
Worked-case data: You have 60 seconds and two candidate minerals. Plan an evidence-first response.

A. Teacher names

B. Only specimen prices

C. Confusions and diagnostic distinctions

D. Only one photograph

Correct: C

9. (2 points) A rock can contain…

Unit: Minerals, rocks, and diagnostic evidence.
Worked-case data: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

A. Only one chemical element

B. No geological material

C. Only transparent crystals

D. Multiple minerals

Correct: D

10. (2 points) Same composition, different structure defines…

Unit: Formation environments, crystal systems, and habit.
Worked-case data: A massive specimen lacks visible faces. Does it lack crystalline order?

A. Cementation

B. Polymorphism

C. Sorting

D. Streak

Correct: B

11. (2 points) Olivine contains…

Unit: Silicate structures and mineral groups.
Worked-case data: Why does a fully shared silica framework have SiO_2 rather than SiO_4?

A. Only metal atoms

B. A pure carbonate lattice

C. No silicon

D. Isolated tetrahedra

Correct: D

12. (2 points) The approved way to distinguish halite is…

Unit: Non-silicates and economic geology.
Worked-case data: A dark specimen has reddish-brown streak and weak magnetism. Which iron mineral is favored?

A. Assume all clear crystals are salt

B. Physical evidence without tasting

C. Taste every sample

D. Heat an unknown

Correct: B

13. (2 points) Fractional crystallization requires…

Unit: Igneous processes, textures, and Bowen's series.
Worked-case data: Granite and rhyolite have similar broad composition but different textures. Explain.

A. Only a color change

B. No crystals

C. Sediment compaction

D. Separation of crystals from melt

Correct: D

14. (2 points) Cementation is…

Unit: Sedimentary processes, grains, and fossils.
Worked-case data: A rock contains rounded gravel clasts in cement. Classify and constrain the claim.

A. Only breaking crystals

B. Binding grains by precipitated material

C. Moving grains downstream

D. Melting a whole rock

Correct: B

15. (2 points) The original rock is the…

Unit: Metamorphism, stress, and mineral assemblages.
Worked-case data: A carbonate-rich rock near an intrusion recrystallizes without melting. Propose an interpretation.

A. Streak

B. Ore grade

C. Unit cell

D. Protolith

Correct: D

16. (2 points) An error log should record…

Unit: Identification strategy, references, and timed practice.
Worked-case data: You have 60 seconds and two candidate minerals. Plan an evidence-first response.

A. Only the answer letter

B. The failed reasoning and correction

C. Only total score

D. Only elapsed time

Correct: B

17. (2 points) The best identification uses…

Unit: Minerals, rocks, and diagnostic evidence.
Worked-case data: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

A. One nickname

B. Only specimen size

C. Several diagnostic observations

D. Color alone

Correct: C

18. (2 points) Acicular describes…

Unit: Formation environments, crystal systems, and habit.
Worked-case data: A massive specimen lacks visible faces. Does it lack crystalline order?

A. Needlelike habit

B. One unique species

C. A hardness value

D. A chemical reaction

Correct: A